

binningverdict: a unified decision system for logarithmic versus linear binning of small-angle neutron scattering data

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Abstract

We present **binningverdict**, an open-source Python package for quantitatively selecting logarithmic or linear binning in small-angle neutron scattering (SANS) data reduction. The framework combines correlation-aware variance–bias optimization with Fisher-information analysis in a computationally efficient workflow operating directly on measured (Q, I, σ) profiles. Central to the method is a closed-form reconstruction-error ratio,

$$R_{\text{MSE}} = E_{\log}^*/E_L^*,$$

obtained by extending the Freedman–Diaconis mean-squared-error decomposition to Q -dependent bin widths. The resulting expression depends only on four elementary moments of the data and is entirely model-free. An optional Fisher-information module evaluates finite- N parameter-estimation precision and demonstrates asymptotic scheme-independence between logarithmic and linear discretizations. The package provides convergence diagnostics, batch-processing capability, and millisecond-scale execution using only NumPy. Benchmarks on canonical SANS models, together with applications to measured EQ-SANS datasets, demonstrate robust performance and establish the framework as a practical tool for automated SANS reduction and preprocessing workflows.

PROGRAM SUMMARY

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Program Title: binningverdict

CPC Library link to program files: (to be added by Technical Editor)

Developer's repository link: <https://github.com/cw-do/binningverdict>

Licensing provisions: MIT

Programming language: Python (≥ 3.8)

External routines/libraries: NumPy (≥ 1.20); SciPy (optional, for cylinder form factor in examples)

Nature of problem: Reducing a SANS measurement to an intensity profile $I(Q)$ requires choosing how to discretize the momentum-transfer axis. The most common choice between logarithmic and linear bin spacing has historically been made by visual convention rather than by quantitative criterion. The literature on SAS data information content (Shannon channels [1], Fisher information [2], etc.) characterizes the information ceiling of a measurement but does not recommend a specific binning scheme for a given profile. There is therefore a need for a single, model-free, computationally cheap procedure that takes the measured (Q, I, σ) profile as input and returns a quantitative recommendation.

Solution method: The Freedman–Diaconis variance–bias decomposition is generalized to a Q -dependent bin width $h(Q)$, the resulting reconstruction mean-squared error is evaluated under linear and logarithmic schemes at matched bin count, and the ratio $R_{\text{MSE}} = E_{\log}^*/E_L^*$ is obtained in closed form as $R_{\text{MSE}} = (J_1^\sigma/\bar{\sigma}^2)^{2/3}(J_2/\bar{\beta})^{1/3}$ with four moments computed by trapezoidal quadrature on the supplied input grid. A self-consistency check on a sub-sampled grid flags inputs whose verdict is not yet stable. An optional Fisher branch evaluates a finite- N ratio $F_{\log}/F_L$ when a parametric model is supplied.

Additional comments: The package is single-file, depends only on NumPy, and runs in a few milliseconds on a typical SANS profile. The repository contains worked examples for canonical SANS models, the Fisher branch, and batch analysis of measured profiles, plus a unit test suite.

1. Introduction

The discretization of the momentum-transfer axis is a fundamental step in small-angle neutron scattering (SANS) data reduction because it directly affects reconstruction accuracy, parameter estimation, and downstream analysis workflows [3]. Despite its practical importance, the common choice between logarithmic and linear binning has historically been guided more by visual convention than by quantitative optimization. Over the past four

decades, several information-theoretic frameworks have characterized the information content of SANS measurements from complementary perspectives. Damaschun *et al.* [4] and Moore [5] introduced the Shannon-channel framework, identifying the number of Shannon channels,

$$N_S = D_{\max}(Q_{\max} - Q_{\min})/\pi,$$

as a sampling-theoretic upper bound on the number of independent intensities recoverable from a scattering profile of finite real-space extent $D_{\max}$, measured over the interval $[Q_{\min}, Q_{\max}]$. Taupin and Luzzati [6] refined the framework for solution scattering. Vestergaard and Hansen [7] introduced the “number of good parameters” N_p as a noise-aware refinement based on Bayesian indirect Fourier transformation, while Rambo and Tainer [8] developed the volume-of-correlation metric V_C . Pedersen *et al.* [9] formulated the problem in terms of Fisher information [2],

$$\mathbb{E}[\mathcal{I}_{ab}(\theta)] \propto \sum_j \frac{\int_{\tilde{Q}_j}^{\tilde{Q}_{j+1}} \partial_a I dQ \int_{\tilde{Q}_j}^{\tilde{Q}_{j+1}} \partial_b I dQ}{\sigma_j^2}, \quad (1)$$

and observed that optimal binning should adapt to regions where the scattering profile varies rapidly. However, they also noted that a practical implementation of Fisher-optimal discretization would be computationally difficult, and the problem was left unresolved. More recent work in the bioSAXS community [10] extended Shannon-based ideas to oversampling-aware fitting strategies, but the specific computational question of logarithmic versus linear sampling of the Q -axis remains unanswered.

In recent work [11, 12], we placed bin-width optimization on a quantitative basis by adapting the Freedman–Diaconis (FD) [13] variance–bias decomposition to SANS data and extending it with Gaussian-process inference to incorporate inter-bin correlations. These studies established scaling relations for optimal bin width under fixed discretization schemes but did not address comparisons between logarithmic and linear sampling. Related work on Bonse–Hart USANS reduction [14] likewise identified the existence of both logarithmic and linear stepping geometries without providing a quantitative selection criterion. The absence of a computationally efficient decision framework motivates the present work.

Here we present **binningverdict**, an open-source Python package that quantitatively determines whether logarithmic, linear, or effectively equivalent binning is preferred for a measured (Q, I, σ) profile. The workflow

computes a closed-form reconstruction-error ratio R_{MSE} , together with an optional Fisher-information ratio R_F , and returns a direct binning recommendation. The reconstruction metric extends the FD variance–bias framework to Q -dependent bin widths and reduces the comparison between logarithmic and linear schemes to four elementary moments of the measured profile. The Fisher-information branch further proves asymptotic scheme-independence and characterizes finite- N corrections relevant to practical parameter estimation. The package is implemented entirely in Python using NumPy, supports batch processing and convergence diagnostics, and executes in milliseconds for typical SANS profiles.

The remainder of the paper is organized as follows. Section 2 derives the reconstruction and Fisher-information criteria. Section 3 presents the workflow algorithm and software interface. Section 4 benchmarks the method on representative SANS models and examines the influence of Q -window selection and measurement protocol. Section 5 applies the workflow to measured EQ-SANS datasets. Section 6 relates the present framework to previous information-theoretic approaches, and Section 7 discusses implications for automated SANS reduction workflows. The results show that power-law scattering systematically favors logarithmic binning, whereas non-power-law profiles at wide- Q windows typical of extended-range instruments such as EQ-SANS [15] generally favor linear discretization, with both schemes becoming asymptotically equivalent for parameter estimation at sufficiently large bin counts.

2. Theory

2.1. Reconstruction MSE for Q -dependent bin width

The Freedman–Diaconis decomposition for SANS, derived in Ref. [11] and refined in Ref. [12], expresses the total reconstruction mean-squared error of a binned estimator $I_h(Q)$ as the sum of counting noise and within-bin aliasing distortion. The key observation we exploit here is that this decomposition is local in Q . Applied to a non-uniform partition with local bin width $h(Q)$, the pointwise error density of Refs. [11, 12] generalizes to

$$\varepsilon(Q; h) = \frac{\sigma^2(Q)}{h(Q)} + \frac{h(Q)^2}{12} [I'(Q)]^2 + \mathcal{O}(h^4), \quad (2)$$

which reduces to the FD result of [11] when h is constant and recovers the Poisson limit when $\sigma^2 \propto I$. The total MSE normalized by the window length

$L = Q_{\max} - Q_{\min}$ is $E = L^{-1} \int \varepsilon(Q; h) dQ$.

2.2. Two canonical schemes and the closed-form ratio

We compare two schemes at fixed total bin count N . The linear scheme uses constant width $h_L = L/N$. The logarithmic scheme uses Q -dependent width $h_{\log}(Q) = Q\Lambda/N$, where $\Lambda = \ln(Q_{\max}/Q_{\min})$.

Each scheme is substituted into the error density Eq. (2). The total MSE in either scheme then takes the familiar variance–bias form $E = \alpha/h + \beta h^2$, with coefficients α and β that depend on the scheme. Minimizing over h inside each scheme gives an optimal error $E^* \propto \alpha^{2/3}\beta^{1/3}$. The ratio of the two optimal errors then has a remarkable property. The window length L , the logarithmic span Λ , and the overall numerical prefactor all cancel out exactly. What remains is a closed-form expression in four moments of the measured profile,

$$R_{\text{MSE}} \equiv \frac{E_{\log}^*}{E_L^*} = \left(\frac{J_1^\sigma}{\sigma^2} \right)^{2/3} \left(\frac{J_2}{\bar{\beta}} \right)^{1/3} \quad (3)$$

with the four moments defined as

$$\begin{aligned} \overline{\sigma^2} &\equiv \frac{1}{L} \int \sigma^2 dQ, & J_1^\sigma &\equiv \frac{1}{L} \int \frac{\sigma^2}{Q} dQ, \\ \bar{\beta} &\equiv \frac{1}{L} \int (I')^2 dQ, & J_2 &\equiv \frac{1}{L} \int Q^2 (I')^2 dQ. \end{aligned} \quad (4)$$

A step-by-step derivation is given in Appendix Appendix A. $R_{\text{MSE}} < 1$ favors log-binning. $R_{\text{MSE}} > 1$ favors linear binning. The expression is model-free, scale-invariant within a fixed window, and reduces to a window-independent constant for power-law $I(Q) \propto Q^{-\alpha}$ (Sec. 4.1).

2.3. Fisher information under the two schemes

Pedersen *et al.* [9] derived the per-bin Fisher information contribution Eq. (1) but did not analyze its dependence on the binning scheme. We carry out that analysis now. Under Poisson counting the expected count in bin k is $N_k \propto I(Q_k)h_k$, and the Fisher information for any structural parameter θ becomes a Riemann sum [16]

$$F(\theta) \propto \sum_{k=1}^N \frac{[\partial_\theta I(Q_k)]^2}{I(Q_k)} h_k. \quad (5)$$

In the continuum limit this converges to $F_\infty = \int [\partial_\theta I]^2 / I dQ$, which depends only on the integrand and not on how Q is discretized. Therefore

$$\boxed{\lim_{N \rightarrow \infty} \frac{F_{\log}(\theta)}{F_L(\theta)} = 1 \quad \text{for any } \theta.} \quad (6)$$

The structural difference between the MSE and Fisher cases can be made explicit by comparing the per-bin contributions side by side. The reconstruction MSE in bin k centered at Q_k is, from Eq. (2) integrated across the bin,

$$\Delta E_k \simeq \underbrace{\frac{\sigma^2(Q_k)}{h_k}}_{\text{counting}} + \underbrace{\frac{h_k^2}{12} [I'(Q_k)]^2}_{\text{aliasing}} \equiv \frac{\alpha_k}{h_k} + \beta_k h_k^2, \quad (7)$$

which contains the bin width h_k in two non-cancelling roles. The counting term scales as h_k^{-1} , so making bins narrower (more bins total) reduces per-bin variance. The aliasing term scales as h_k^2 , so making bins wider blurs the underlying $I(Q)$. The two roles compete, and the optimum at fixed total bin count depends on the scheme.

By contrast, the per-bin Fisher contribution under Poisson counting is

$$\Delta F_k = \frac{[\partial_\theta I(Q_k)]^2}{\text{Var}(N_k)} (h_k)^2 = \frac{[\partial_\theta I(Q_k)]^2}{I(Q_k) h_k} h_k^2 = \frac{[\partial_\theta I(Q_k)]^2}{I(Q_k)} h_k. \quad (8)$$

The second equality uses $\text{Var}(N_k) \propto I(Q_k) h_k$ for Poisson counts. The third equality combines $h_k^2/h_k = h_k$, so a single power of h_k survives at the end.

This surviving factor of h_k deserves a closer look. It does not act as a free knob that the experimenter can tune in either direction. It acts as the local width of the partition that approximates the underlying integral. Recall how a Riemann sum is constructed. To approximate $\int g(Q) dQ$ on a grid $\{Q_k\}$, one writes $\sum_k g(Q_k) \Delta Q_k$, where ΔQ_k is the local spacing of the grid. As the grid is refined, ΔQ_k shrinks at every k and the sum converges to the integral. Crucially, the value of the limiting integral does not depend on how the grid was refined. A uniform refinement and a non-uniform refinement of the same window give the same answer.

Eq. (8) has exactly the structure of a Riemann sum. The quantity $[\partial_\theta I(Q_k)]^2 / I(Q_k)$ plays the role of the integrand, while h_k represents the local interval width ΔQ_k . Summing over k and taking the limit $N \rightarrow \infty$ therefore yields

$$F_\infty = \int \frac{[\partial_\theta I(Q)]^2}{I(Q)} dQ.$$

Whether the discretization is logarithmic or linear, both schemes converge to the same continuum integral. This is the content of Eq. (6).

The MSE in Eq. (7) has no analogous structure. The counting term has h_k in the denominator, the aliasing term has h_k squared in the numerator, and these two scalings cannot be combined into a single Riemann measure. Refining the grid does not push the MSE into a partition-independent limit. Instead, it traces out a variance–bias curve whose minimum location and value depend on how the grid is laid down.

In short, the two opposing effects of refining the binning—smaller variance per bin against more bins—trade off against each other in the Fisher case and exactly cancel in the continuum limit. In the MSE case the two effects come from physically distinct sources, namely statistical noise and within-bin signal blur, and their competition does not vanish. Reconstruction sees the scheme. Parameter information does not, asymptotically.

The finite- N residual to Eq. (6) is dictated by the trapezoidal-rule convergence rate of each scheme on the integrand $(\partial_\theta I)^2/I$. This is characterized numerically in Sec. 4.6.

3. Workflow and software

The theory of Sec. 2 is converted into an experimental workflow as Algorithm 1.

Algorithm 1 Log-vs-linear binning recommendation

- 1: **Input:** measured profile (Q, I, σ) ; optional parametric model $I(Q; \theta), \partial_\theta I(Q; \theta)$ at current best θ ; anticipated bin count N .
 - 2: Estimate $I'(Q)$ from (Q, I) by central finite differences.
 - 3: Compute the four moments of Eq. (4) by trapezoidal quadrature.
 - 4: Evaluate R_{MSE} via Eq. (3).
 - 5: Repeat steps 2–4 on a sub-sampled grid (every other point); flag a convergence warning if the two values disagree by more than 10%.
 - 6: **if** parametric model provided **then**
 - 7: Compute $F_L, F_{\log}$ via Eq. (5) at N bins.
 - 8: $R_F \leftarrow F_{\log}/F_L$.
 - 9: **end if**
 - 10: **Verdict:** “log” if $R_{\text{MSE}} < 0.90$; “linear” if $R_{\text{MSE}} > 1.10$; “tied” otherwise.
-

The workflow is implemented in the open-source Python package `binningverdict`,

Table 1: R_{MSE} for power-law $I(Q) = Q^{-\alpha}$.

α	R_{MSE}
1.0	0.426
2.0	0.748
3.0	0.854
4.0	0.898

distributed under the MIT license at <https://github.com/<your-username>/binningverdict>. The package depends only on NumPy. The user-facing API consists of a single function:

```
from binningverdict import analyze_binning
result = analyze_binning(Q, I, err=sigma)
print(result['verdict'])      # 'linear' / 'log' / 'tied'
print(result['R_MSE'])       # numerical ratio
```

A parametric model can optionally be supplied to also evaluate the Fisher ratio. The result dictionary additionally reports the sub-sampled R_{MSE} used for self-consistency checking, a Boolean `converged` flag, and a human-readable rationale string. The repository contains worked examples for canonical models, the Fisher branch, and batch analysis of measured profiles, plus a unit test suite.

4. Benchmark results

We validate the workflow on representative SANS profiles. Unless noted, the reference window is $[Q_{\min}, Q_{\max}] = [10^{-3}, 0.5] \text{ \AA}^{-1}$, sampled at 10^4 log-spaced points.

4.1. Power-law scattering

For $I(Q) = AQ^{-\alpha}$, the profile is scale-invariant and R_{MSE} is window-independent. Table 1 reports the Poisson-limit values; $R_{\text{MSE}} < 1$ at every α , with $R_{\text{MSE}} \rightarrow 1$ from below as $\alpha \rightarrow \infty$. Power-law scattering is the unique scale-invariant regime in which log-binning is strictly preferred at any window.

Table 2: R_{MSE} for non-power-law profiles at the reference window.

Family	Parameters	R_{MSE}
Guinier	$R_g = 30 \text{ \AA}$	2.42
Guinier	$R_g = 100 \text{ \AA}$	1.93
Sphere	$R = 30 \text{ \AA}$	2.57
Sphere	$R = 100 \text{ \AA}$	2.07
Cylinder	$R=20, L=200 \text{ \AA}$	1.86
Cylinder	$R=50, L=1000 \text{ \AA}$	1.15
Lorentzian peak	$Q^*=0.05, \Gamma=0.01$	1.16
Teubner–Strey	$(1, -100, 10^5)$	2.39

Table 3: R_{MSE} at three representative low- Q cutoffs, $Q_{\text{max}} = 0.5 \text{ \AA}^{-1}$.

Model	$Q_{\text{min}}=0.002$	$Q_{\text{min}}=0.007$	$Q_{\text{min}}=0.05$
Power law $\alpha = 2$	0.75	0.75	0.80
Power law $\alpha = 4$	0.90	0.90	0.90
Guinier $R_g = 30$	2.14	1.63	1.00
Sphere $R = 30$	2.28	1.76	1.05
Cylinder $R=20, L=200$	1.62	1.18	0.98
Lorentzian peak	1.13	1.07	0.94
Teubner–Strey	2.14	1.67	0.95

4.2. Non-power-law profiles

Table 2 collects R_{MSE} values for Guinier, sphere, cylinder, Lorentzian peak, and Teubner–Strey profiles at the reference window. All are above unity, indicating linear-bin advantage. The magnitude is largest for compact objects such as spheres, which concentrate intensity at low Q .

4.3. Dependence on the measurement window

Figure 1 maps out how R_{MSE} varies with Q_{min} and Q_{max} . Power-law curves are strictly horizontal, confirming scale invariance. All non-power-law profiles show $R_{\text{MSE}} \rightarrow 1$ as the window narrows; the linear-bin advantage is largest at extended- Q operation.

Table 3 summarizes R_{MSE} at three representative low- Q cutoffs.

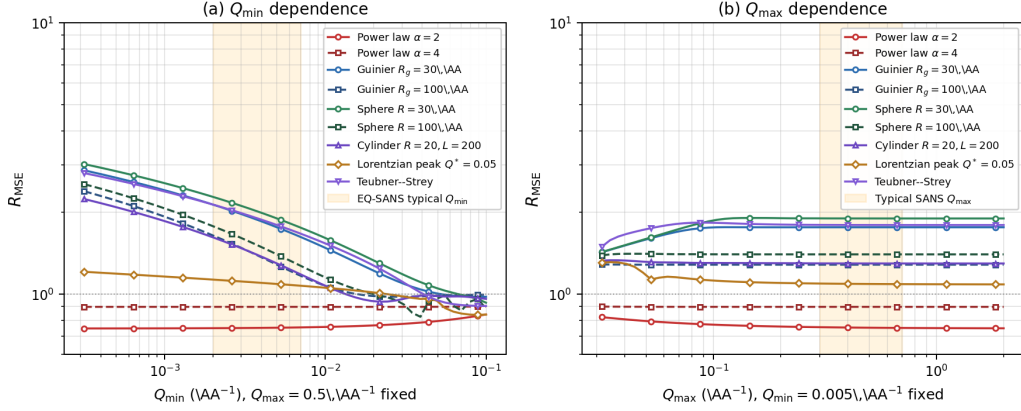


Figure 1: Window dependence of R_{MSE} across nine representative profiles. (a) $Q_{\min}$ sweep at fixed $Q_{\max} = 0.5 \text{ \AA}^{-1}$; shaded band marks typical EQ-SANS low- Q range. (b) $Q_{\max}$ sweep at fixed $Q_{\min} = 0.005 \text{ \AA}^{-1}$. Power-law profiles are scale-invariant (horizontal). All non-power-law profiles approach $R_{\text{MSE}} = 1$ as the window narrows.

Table 4: Sensitivity of R_{MSE} to the measurement noise model. Guinier profile, $R_g = 30 \text{ \AA}$, reference window.

Noise model $\sigma^2(Q)$	R_{MSE}
Poisson, $\sigma^2 = I(Q)$ (baseline)	2.14
Poisson + high- Q background floor	2.13
Q-dependent counting time, low- Q weighted $\times 10$	1.31
Constant σ (background-dominated)	0.67

4.4. Effect of measurement protocol

The generalized form (3) uses measured $\sigma(Q)$ rather than the Poisson proxy $\sigma^2 \propto I$. Table 4 shows how R_{MSE} for a fixed Guinier profile responds to four noise scenarios. The verdict is stable for signal-dominated measurements, weakens with low- Q -weighted counting time, and can flip under fully background-dominated acquisition where σ^2 becomes constant.

4.5. Input quadrature convergence

Figure 2 characterizes how the workflow’s verdict converges with the input bin count N_{in} . For non-power-law profiles (Guinier, sphere, Lorentzian, Teubner–Strey) both schemes agree to within 3% by $N_{\text{in}} = 50$ and the verdict is stable across input schemes. Power-law profiles are the slowest-converging

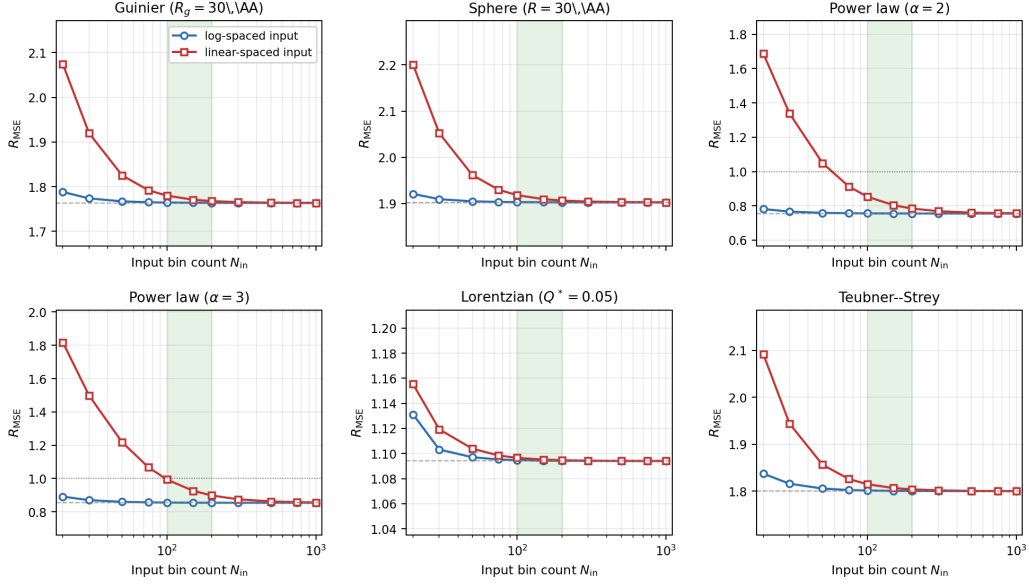


Figure 2: Convergence of the R_{MSE} estimator with input bin count N_{in} . Both schemes converge to the same limit; for power-law profiles the convergence is slower in the linear-input branch.

Table 5: Verdict consistency between log- and linear-spaced input schemes at $N_{\text{in}} = 100$. Verdicts: “L”=linear, “log”=logarithmic, “~”=tied. R_{MSE} values shown in parentheses.

Model	log input	linear input	Consistent?
Guinier $R_g=30$	L (1.76)	L (1.78)	yes
Sphere $R=30$	L (1.90)	L (1.92)	yes
Lorentzian peak	~(1.10)	~(1.10)	yes
Teubner–Strey	L (1.80)	L (1.81)	yes
Power law $\alpha=2$	log (0.76)	~(0.85)	no
Power law $\alpha=3$	log (0.86)	~(0.99)	no

case: verdicts can disagree between the two input schemes at modest N_{in} (Table 5), and the self-consistency check in Algorithm 1 flags this condition automatically.

Two practical recommendations follow. First, supply log-spaced input where possible, since convergence is faster. Second, when the convergence diagnostic flags a power-law-like profile, recompute on a denser input grid by

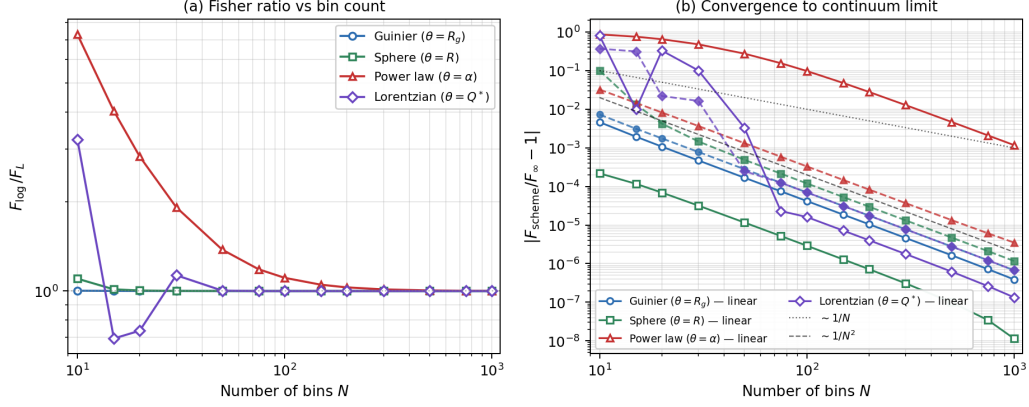


Figure 3: Finite- N Fisher information under linear and log binning. (a) Ratio $F_{\log}/F_L$ versus N for four representative models. All curves approach unity in the large- N limit, confirming Eq. (6). (b) Convergence of each scheme to the continuum Fisher integral. Solid lines mark the linear scheme, dashed lines mark the log scheme. Reference slopes $1/N$ and $1/N^2$ indicate expected Riemann-sum convergence.

supplying a less aggressively rebinned profile, and continue until the verdict stabilizes.

4.6. Fisher information: asymptotic equivalence

Figure 3 shows the finite- N Fisher analysis. Panel (a) plots $F_{\log}/F_L$ versus N for four representative models. All curves approach unity as N grows, confirming Eq. (6). Panel (b) shows the absolute deviation from the continuum limit. For Guinier, sphere, and Lorentzian profiles, both schemes reach the continuum limit to within 0.1% by $N \approx 100$; parameter precision is scheme-independent at any realistic SANS bin count. Power-law scattering is the exception: $F_{\log}/F_L \approx 3$ at $N = 20$ and convergence to within 1% requires $N \gtrsim 500$.

5. Application to measured EQ-SANS data

We close the validation by applying the workflow to five measured EQ-SANS profiles spanning distinct soft-matter and porous-material classes (Fig. 4, Table 6). The first four samples—representing distinct soft-matter classes—all return a linear verdict at the wide- Q EQ-SANS window in which they were measured. R_{MSE} values cover a substantial range (1.24–2.57), with the smallest advantage for D100, whose narrower window $[0.007, 0.12] \text{ \AA}^{-1}$

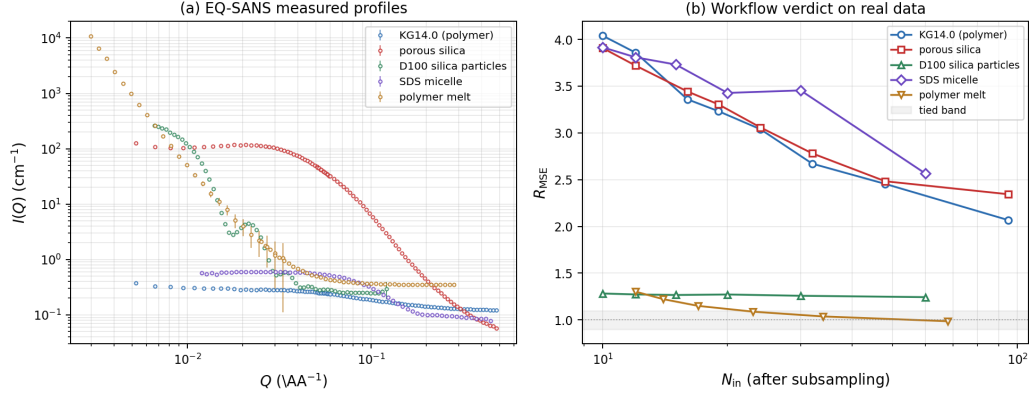


Figure 4: Workflow applied to five measured EQ-SANS profiles. (a) Measured intensity profiles. (b) R_{MSE} as a function of input bin count N_{in} , obtained by subsampling each profile by progressively larger strides. Shaded band marks the tied verdict region.

Table 6: Workflow verdict on five measured EQ-SANS profiles at full input resolution.

Sample	N_{in}	Q range ($\text{\AA}^{-1}$)	R_{MSE}	Verdict
KG14.0 (polymer)	95	[0.005, 0.48]	2.07	linear
porous silica	95	[0.005, 0.48]	2.34	linear
D100 silica particles	60	[0.007, 0.12]	1.24	linear
SDS micelle	60	[0.012, 0.45]	2.57	linear
polymer melt	68	[0.003, 0.28]	0.99	tied

moves the verdict toward the tied regime in line with the trend of Fig. 1. SDS micelles, where peak-like correlation features dominate, give the largest linear-bin advantage. These outcomes confirm the central prediction: at extended- Q operation typical of EQ-SANS, non-power-law profiles benefit measurably from linear binning.

The polymer melt is qualitatively different and illustrates the workflow’s behavior on a sample with mixed character. At low Q the profile follows a near-Porod $I \propto Q^{-4}$ decay over almost two decades, while at $Q \gtrsim 0.05 \text{ \AA}^{-1}$ it transitions to an essentially flat plateau. Restricted analyses on the two regions separately give $R_{\text{MSE}} \approx 0.99$ on the Porod region (consistent with the $\alpha \rightarrow 4$ limit of Table 1) and $R_{\text{MSE}} \approx 0.71$ on the plateau region (where flat $I(Q)$ favors log-binning). Combined over the full window, the two contributions balance to a tied verdict, $R_{\text{MSE}} = 0.99$. The subsampling curve in

Table 7: Relation of the present work to prior frameworks.

Reference	Quantity	Question addressed
Damaschun Moore [5]	[4], Shannon channels N_S	Maximum independent intensities for given $D_{\max}$
Taupin & Luzzati [6]	Refined N_S	Solution-scattering specifics
Vestergaard Hansen [7]	& Number of good params N_p	Noise-aware effective dimensionality
Rambo & Tainer [8]	Volume of correlation V_C	Mass and resolution from oversampled data
Pedersen [9]	Fisher matrix Eq. (1)	Optimal binning in principle (not pursued)
Tung <i>et al.</i> [11, 12]	FD/GP optimal h at fixed scheme	Bin- <i>width</i> selection within a scheme
Present work	R_{MSE} , asymptotic $F_{\log}/F_L$	Cross- <i>scheme</i> comparison: log vs. linear

Fig. 4(b) further shows that this verdict is itself N_{in} -sensitive: at $N_{\text{in}} \lesssim 20$ the verdict drifts into the linear regime, and the convergence diagnostic built into the workflow flags this case automatically.

6. Relation to prior information-theoretic frameworks

The present work synthesizes rather than supersedes earlier analyses of SAS information content. Table 7 summarizes.

The Shannon-channel framework [1, 4, 5, 6] sets a sampling-theoretic upper bound on the number of independent intensities recoverable from a profile of finite real-space extent, but is silent on how those intensities should be distributed in Q . N_p [7] and V_C [8] refine the information bound by accounting for noise and correlation, but again make no recommendation about the distribution of bins. Pedersen *et al.* [9] did pose the binning question through the Fisher matrix Eq. (1) but explicitly noted that translating it into a practical algorithm would be “somewhat complicated” and left it as an open direction. The present work follows that direction by restricting attention to the two practically relevant canonical schemes (log and linear) rather than the full optimization, which permits the closed form Eq. (3) and removes the

model dependence in the MSE branch. The Fisher branch sharpens Peder-
sen’s observation by showing asymptotic scheme-independence, a conclusion
that was not articulated in [9].

Our recent FD and GP-based analyses [11, 12] provide the variance–bias
decomposition of which Eq. (2) is the Q -dependent extension. Those works
answered “how wide should the bin be?” The present work answers “how
should the bins be distributed?” The two layers compose. The user picks
the scheme by R_{MSE} and the within-scheme bin count by FD/GP.

7. Discussion

The unified verdict is window-dependent and operationally simple. Power-
law profiles favor log-binning at every Q window. Non-power-law profiles
favor linear at extended- Q windows characteristic of EQ-SANS-class instru-
ments, with ratios of 1.5–2.3 in the Poisson limit. At narrow windows the
schemes converge to statistical equivalence.

The practical implication splits along the use case. When the reduced
profile $I(Q)$ feeds further reconstruction-stage processing such as multiple-
scattering correction, desmearing of USANS profiles [14], indirect Fourier
transform, Kratky displays, or model-free data releases, the binning choice
has direct consequences. The effect is largest at the wide Q -windows where
modern instruments operate. When the only target is parameter estimation
through structural model fitting, asymptotic scheme-independence makes the
choice immaterial at typical bin counts.

A practical complement to fixed-scheme binning is fully adaptive $h(Q)$,
obtained by pointwise minimization of Eq. (2). This yields

$$h^*(Q) \propto \left[\frac{\sigma^2(Q)}{(I'(Q))^2} \right]^{1/3}.$$

Time-of-flight SANS instruments that store individual neutron events [15]
permit such post-hoc adaptive rebinning. The reduction-stack overhead,
however, makes the present fixed-scheme verdict the most immediately ac-
tionable result. Inter-bin correlation corrections introduced in Ref. [12] mod-
ify the coefficients in Eq. (3) and will be treated in future work.

8. Conclusion

We have presented `binningverdict`, an open-source Python package for
quantitative selection between logarithmic and linear binning in SANS data

reduction workflows. The framework combines a closed-form reconstruction-error estimator, $R_{\text{MSE}} = E_{\log}^*/E_L^*$, with an optional Fisher-information analysis to provide an automated and computationally inexpensive binning recommendation directly from measured (Q, I, σ) profiles. The reconstruction criterion is obtained by extending the Freedman–Diaconis variance–bias decomposition to Q -dependent bin widths, while the Fisher branch establishes asymptotic scheme-independence for parameter estimation.

Benchmark calculations on canonical scattering models and applications to measured EQ-SANS datasets demonstrate that power-law profiles systematically favor logarithmic binning in a scale-invariant manner, whereas non-power-law profiles at the wide Q -windows typical of extended-range instruments generally favor linear discretization. The workflow further includes a self-consistency diagnostic that automatically identifies input grids for which the verdict has not yet converged.

Because the method requires only elementary quadrature and finite-difference operations, it executes in milliseconds using only NumPy and is suitable for integration into automated SANS and USANS reduction pipelines where reproducible discretization criteria are important for downstream analysis and reconstruction accuracy. The package is distributed under the MIT license and provides batch-processing utilities, worked examples, and an optional Fisher-analysis interface through the single function call `analyze_binning(Q, I, sigma)`.

Appendix A. Step-by-step derivation of Eq. (3)

This appendix derives the closed-form ratio $R_{\text{MSE}} = E_{\log}^*/E_L^*$ in detail. We start from the Q -dependent error density

$$\varepsilon(Q; h) = \frac{\sigma^2(Q)}{h(Q)} + \frac{h(Q)^2}{12} [I'(Q)]^2, \quad (\text{A.1})$$

identical to Eq. (2) of the main text. The total MSE normalized by the window length $L = Q_{\text{max}} - Q_{\text{min}}$ is

$$E[h] = \frac{1}{L} \int_{Q_{\text{min}}}^{Q_{\text{max}}} \varepsilon(Q; h) dQ. \quad (\text{A.2})$$

Linear scheme

The linear scheme uses constant bin width $h_L = L/N$. Substituting $h(Q) = h_L$ into Eq. (A.1) and integrating gives

$$E_L(h_L) = \frac{1}{h_L} \overline{\sigma^2} + \frac{h_L^2}{12} \bar{\beta}, \quad (\text{A.3})$$

where the moments

$$\overline{\sigma^2} = \frac{1}{L} \int \sigma^2 dQ, \quad \bar{\beta} = \frac{1}{L} \int (I')^2 dQ \quad (\text{A.4})$$

are simple Q -averages of σ^2 and $(I')^2$.

Logarithmic scheme

The logarithmic scheme uses the Q -dependent width $h_{\log}(Q) = Q\delta$, with $\delta = \Lambda/N$ and $\Lambda = \ln(Q_{\max}/Q_{\min})$. Substituting into Eq. (A.1) gives

$$\varepsilon_{\log}(Q; \delta) = \frac{\sigma^2(Q)}{Q\delta} + \frac{Q^2 \delta^2}{12} [I'(Q)]^2. \quad (\text{A.5})$$

Integrating over Q and dividing by L yields

$$E_{\log}(\delta) = \frac{1}{\delta} J_1^\sigma + \frac{\delta^2}{12} J_2, \quad (\text{A.6})$$

where the new moments

$$J_1^\sigma = \frac{1}{L} \int \frac{\sigma^2}{Q} dQ, \quad J_2 = \frac{1}{L} \int Q^2 (I')^2 dQ \quad (\text{A.7})$$

arise from the explicit factors of Q^{-1} and Q^2 produced by substituting $h_{\log}(Q) = Q\delta$.

Optimal errors in each scheme

Both Eq. (A.3) and Eq. (A.6) have the same generic form

$$E(x) = \frac{a}{x} + b x^2, \quad (\text{A.8})$$

for a width parameter x and positive coefficients a, b . The minimum is found by setting $dE/dx = -a/x^2 + 2bx = 0$, giving

$$x^* = \left(\frac{a}{2b} \right)^{1/3}, \quad E^* = \frac{3}{2^{2/3}} a^{2/3} b^{1/3}. \quad (\text{A.9})$$

Applied to the linear scheme with $a = \overline{\sigma^2}$ and $b = \bar{\beta}/12$ yields

$$h_L^* = \left(\frac{6 \overline{\sigma^2}}{\bar{\beta}} \right)^{1/3}, \quad E_L^* = K \overline{\sigma^2}^{2/3} \bar{\beta}^{1/3}, \quad (\text{A.10})$$

with $K = (3/2^{2/3}) 12^{-1/3}$. Applied to the logarithmic scheme with $a = J_1^\sigma$ and $b = J_2/12$ yields

$$\delta^* = \left(\frac{6 J_1^\sigma}{J_2} \right)^{1/3}, \quad E_{\log}^* = K (J_1^\sigma)^{2/3} J_2^{1/3}. \quad (\text{A.11})$$

Cancellation in the ratio

Forming the ratio $E_{\log}^*/E_L^*$, the prefactor K cancels exactly. The window length L that appears implicitly through the moment definitions also cancels, since both numerator and denominator moments are normalized by L . The logarithmic span Λ does not appear at all in the optimal errors, only in the relation between δ^* and the actual bin count $N = \Lambda/\delta^* = N(J_1^\sigma, J_2, \Lambda)$, and this Λ does not propagate into $E_{\log}^*$ itself. The remaining expression is

$$R_{\text{MSE}} = \frac{E_{\log}^*}{E_L^*} = \left(\frac{J_1^\sigma}{\overline{\sigma^2}} \right)^{2/3} \left(\frac{J_2}{\bar{\beta}} \right)^{1/3}, \quad (\text{A.12})$$

which is Eq. (3) of the main text.

Poisson special case

When $\sigma^2(Q) = c I(Q)$ for a positive constant c (the Poisson limit), the constant c appears in the same way in $\overline{\sigma^2}$ and in J_1^σ , and it cancels in their ratio. The expression then reduces to

$$R_{\text{MSE}}^{\text{Poisson}} = \left(\frac{J_1}{\bar{I}} \right)^{2/3} \left(\frac{J_2}{\bar{\beta}} \right)^{1/3}, \quad (\text{A.13})$$

with $\bar{I} = L^{-1} \int I dQ$ and $J_1 = L^{-1} \int I/Q dQ$. The Poisson-limit form is what we use for all model-generated benchmarks in Sec. 4.

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